ANKITH REDDY SUBHANPURAM

Gainesville, FL, USA | ankithreddy653@gmail.com | +1 (352) 642-5515
ankithreddys.vercel.app | LinkedIn | GitHub

Research Interests

Architecture and training choices for **multimodal foundation models** over language, video and audio: how modality-specific and shared representations should interact, where cross-modal fusion should occur, how attention, routing and modularity affect long-context integration, and how update strategies and diagnostic evaluation explain generalization and capability retention. Prior work: multimodal EHR Transformers, LLM adaptation, controlled MoE interventions, low-resource ASR. Applying for Fall 2027 PhD entry.

Education

University of Florida Gainesville, FL | Aug 2024–May 2026

M.S., **Artificial Intelligence Systems**, Dept. of Engineering Education | GPA 3.90/4.00

Vignana Bharathi Institute of Technology Hyderabad, India | Aug 2020–May 2024

B.Tech., **Computer Science and Engineering** (AI and Machine Learning) | CGPA 8.76/10.00

Research Appointment

UF Intelligent Clinical Care Center (IC3), PRISMAp Lab Gainesville, FL | Aug 2025–Present

ML Researcher (full-time since May 2026); previously Research Assistant.
PI: Dr. Azra Bihorac | Supervisors: Dr. Zhenhong Hu, Dr. Akshith Ullal

Manuscripts (Working Titles; Unpublished)

- **My Surgical Risk: Continuous, Dynamic, and Explainable Risk Estimation of Surgical Complications Across the Perioperative Continuum.** Manuscript complete; preprint in preparation.
- **PRISMA-ICU: Physician-Curated Reasoning and Instruction-Tuned Specialized Model for Intensive Care Units.** Automated evaluation complete; clinical evaluation ongoing.

Research Experience

MySurgeryRisk: Multimodal, Multitask Transformer for Perioperative Risk *Collaborative; manuscript complete*

- Co-developed a ∼1.7M-parameter multimodal, multitask Transformer predicting outcomes at preoperative, end-of-surgery and daily postoperative decision points from static features, irregular time series and clinical notes.
- Contributed to preprocessing and token design: 128-D typed variable–value–relative-time tokens, with raw timestamps pre- and intraoperatively and six-hour summary bins postoperatively.
- Worked on long-context handling: a 4,096-token budget that keeps recent observations and pools older ones by variable. The shorter context outperformed ∼8,096 tokens on training efficiency and AUROC/AUPRC.
- Contributed to note integration and fusion: frozen BioClinical ModernBERT chunks (1,024 tokens, 256 overlap) projected into the token space; task-token self-attention over structured tokens, then task-token cross-attention into note chunks. Participated in implementation and training.

Independent MoE Fine-Tuning Investigation *Independent design; evaluation ongoing*

- Domain SFT of **gpt-oss** MoE models (see PRISMA-ICU) appeared to degrade structured special-token formatting and tool calling. Designed controlled experiments to ask which parameter groups govern capability retention, rather than assuming a cause.
- Partitioned weights into shared, router and expert groups and specified all $2^3 = 8$ train/freeze combinations (seven trained plus a frozen reference). Trained each on ICU-domain and ∼200K-example UltraChat general SFT: **14 runs, 112 checkpoints** (∼500–19,000 steps), to compare early vs. late adaptation, domain vs. general data, and individual vs. joint group updates.
- Evaluating checkpoint trajectories on MMLU-Pro, BBH, GSM8K, TruthfulQA, XSTest, HarmBench, BBQ, IFEval and BFCL V3 (reasoning, truthfulness, safety, bias, instruction following, tool calling). Training and generation complete.

PRISMA-ICU: Conversational LLM Adaptation *Collaborative; clinical evaluation ongoing*

- Built SFT data from a clinician-designed framework (domains, patient/provider views, four task formats; ∼162 diseases, ∼51 procedures): **97,000+** structured questions, rewritten into natural phrasing and answered via frontier-model prompting (Gemini 2.5 Pro), giving 97,000+ accepted QA pairs (∼80/10/10 split). Clinicians supplied templates, not answers.
- Implemented a three-judge majority-vote quality framework (Gemini 3.1 Pro, Claude 4.5 Sonnet, GPT-5.4) with a rubric covering medical accuracy, safety, completeness, empathy and ethical principles, plus a separate three-judge pairwise framework for comparing post-SFT models.
- Fine-tuned Llama 3.3 70B, gpt-oss 20B (MoE) and Gemma 4 31B (dense) and benchmarked them against additional base models; Gemma performed best of the three in automatic comparisons.
- Running a clinician comparison of base vs. fine-tuned Gemma on 100 held-out SICU examples; inter-rater agreement (Cohen's kappa) planned.

Low-Resource Bengali Speech Recognition, DRDO CAIR *Research Intern | Oct 2023–Mar 2024*

- Took on speech recognition for **low-resource Indian languages** (Bengali, Assamese), where labeled speech is scarce, working with a proprietary, non-public Bengali corpus of speech and transcripts. Mentor: Naval Gupta.
- Surveyed competing ASR architectures and chose **wav2vec 2.0**, fine-tuning a pretrained self-supervised speech encoder on Bengali rather than training from scratch.
- Made the **output vocabulary the experimental variable**: fine-tuned with character-level vs. byte-pair-encoding (BPE) targets and found that **BPE gave better recognition** in the tested setup.
- Why it matters to my research: first-hand evidence that what a pretrained audio model predicts over changes how well it adapts in a low-resource regime. It is the same representation question I now ask of typed EHR tokens and MoE parameter groups, and the root of my interest in language–audio–video models.

Applied AI Experience

Oelrich Construction: AI Engineer, Co-op Aug 2025–May 2026

- Built a permissions-aware RAG assistant over construction documents, meetings and media with role-based access (administrators, employees, leadership); supported cloud deployment.

Scholarship Auditions: AI Engineer, Intern May–Aug 2025

- Developed a conversational RAG assistant for a music scholarship and networking platform; contributed to the deployed chatbot and cloud services.

Teaching & Academic Communication

- **Teaching Assistant**, Foundations of Machine Learning, VBIT (two semesters, 2022–2023): course materials and substitute lectures under Dr. Shirisha Kasireddy.
- **Invited Speaker**, VBIT AI Days, "ML Edge: Innovation and Inspire" (2023): seminar on supervised and unsupervised learning for ∼500 students.

Technical Skills

Research and modeling: Python, PyTorch, TensorFlow, Hugging Face Transformers, PEFT/LoRA, TRL, DeepSpeed, multimodal fusion, SFT, MoE, LLM evaluation, RAG, low-resource ASR.

Systems: vLLM, CUDA, Slurm, Docker, Git, SQL, AWS, Azure.